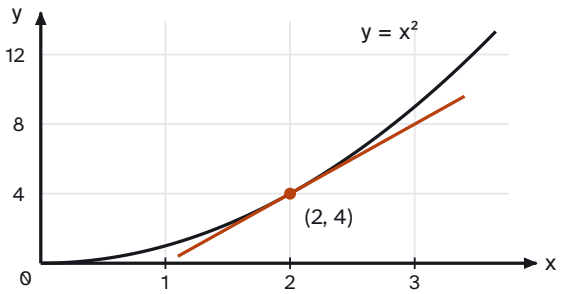
C1

Calculus I

Understand the idea. Build the method. Earn the marks.

FIG. 04 / CALCULUS

A tangent measures the local rate of change



At x = 2

Slope = 2x = 4

Tangent equation:
y - 4 = 4(x - 2)

Takeaway: use coordinate changes for slope; the axes have different scales.

Document	Calculus I
Level / lessons	3 / 12
Edition	November 2026 / Bench Sheet
Pages	65

Start here

Begin with one lesson and a pencil. Follow the stages below, then practise without notes.

1. Read the Idea and describe it in your own words before looking at the rule.
2. Check what every symbol means and when the rule applies.
3. Cover the worked steps. Predict the next step, then compare.
4. Use the working grid to attempt Your turn. Check the answers in the lesson review section.
5. Attempt the subject tests without notes. Use the explanations to identify the exact step you missed.

Read the page at a glance

Rule

$$v = \frac{d}{t}$$

A framed rule is something to select and apply, not just memorise. Here, average speed is distance divided by elapsed time.

Your turn

Blue marks a task for you. Write a prediction or calculation before checking.

Watch out

An orange note marks a common error. Example: do not confuse distance with displacement.

Before you begin

- Have a calculator, pencil and the units used in your course.
- Review basic rearrangement and graph reading when a lesson lists them as pre-requisites.
- Use the formula sheet and symbol glossary for revision; the same letter can have different meanings in different lessons.

Use the linked contents and alphabetical topic index to find a lesson. Marking points are study guidance, not an official marking scheme.

Your study route

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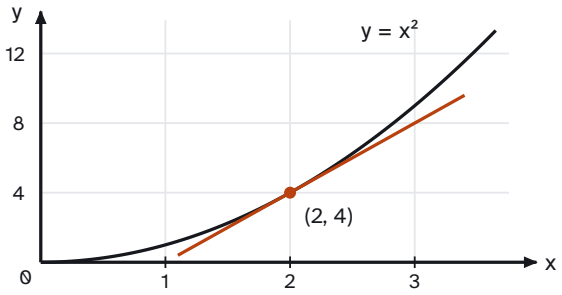
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Calculus I

FIG. 04 / CALCULUS

A tangent measures the local rate of change



At x = 2

Slope = 2x = 4

Tangent equation:
y - 4 = 4(x - 2)

Takeaway: use coordinate changes for slope; the axes have different scales.

MATH 203: derivative rules, implicit/logarithmic/exponential derivatives, L Hopital, graphing and extrema, optimization, Rolle and mean-value theorems, integration and the fundamental theorem. The full lecture syllabus is the main source.

Subject map

MATH 203: derivative rules, implicit/logarithmic/exponential derivatives, L Hopital, graphing and extrema, optimization, Rolle and mean-value theorems, integration and the fundamental theorem. The full lecture syllabus is the main source.

Follow this reading route from foundations to applications. Each line links to its lesson; revisit the stated prerequisites whenever a step feels unfamiliar.

01 Limits, continuity and the derivative definition p.6

02 Basic derivatives and tangent lines p.9

03 Product and quotient rules p.12

04 Chain rule and trigonometric functions p.15

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06 Logarithmic, exponential and inverse-trig derivatives p.22

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12 Definite integrals, signed area and the fundamental theorem p.41

Limits, continuity and the derivative definition

C1 / 01

- Distinguish a limit from a value at the point.
- Find a polynomial derivative from the difference quotient.

You need: Expanding a square; factoring and cancelling a nonzero common factor.

01 Idea

A limit describes nearby values, not necessarily the function value at the point. Continuity requires the function value and the two-sided limit to exist and agree. The derivative is the limiting slope of secant lines and gives instantaneous rate of change.

A secant line measures change across separated inputs. As those inputs approach one another, its slope can approach a single instantaneous slope: the derivative. The increment must remain nonzero while it is a denominator, even though the limiting process eventually removes it.

- Factoring and cancelling the increment can remove the apparent zero denominator.
- The nearby simplified slopes determine the limit, rather than an undefined quotient at zero increment.

02 Rule and symbols

Rule

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

h small nonzero input increment before the limit / same unit as x

$f'(x)$ instantaneous rate of change / output unit per input unit

03 Method

1. Write f(x+h) and subtract the complete original f(x).
2. Expand brackets carefully and collect terms.

3. Factor out h and cancel only while h is nonzero.
4. Take the limit h approaching zero in the simplified expression.

If the question says use the definition, construct the difference quotient instead of citing the power rule.

04 Worked example

For $f(x) = x^2$, the difference quotient becomes $\frac{2xh+h^2}{h} = 2x + h$.

01 For f(x) = x^2, the incremented value is $(x+h)^2$ = x^2 + 2xh + h^2.

02 Subtracting f(x) leaves 2xh + h^2.

03 Dividing by nonzero h gives 2x + h.

04 As h approaches zero, the secant slope approaches 2x, so f′(x) = 2x.

Watch out

Zero over zero is an indeterminate form, not a final answer of zero.

Exam habit

Place brackets around f(x+h) and f(x) before subtraction, especially when either contains negative terms.

05 Your turn

Use the definition to find the derivative of f(x)=3x squared.

If you get stuck, uncover one hint at a time:

1. Write $f(x+h) = 3(x+h)^2$, then subtract the complete original function.
2. Expand and factor h from the numerator of the difference quotient.
3. Cancel nonzero h, then let it approach zero. Keep the coefficient 3 throughout.

06 Check your understanding

Why should h not be set to zero before simplifying the difference quotient?

Why can the increment be cancelled before taking the derivative limit?

1. It is nonzero while approaching zero
2. It equals zero at every stage
3. Any denominator may always be cancelled

Answers and explanations: page 45.

Basic derivatives and tangent lines

C1 / 02

- Differentiate sums and powers correctly.
- Build a tangent line from the original point and derivative slope.

You need: Power rule; substituting an x value; point-slope form of a line.

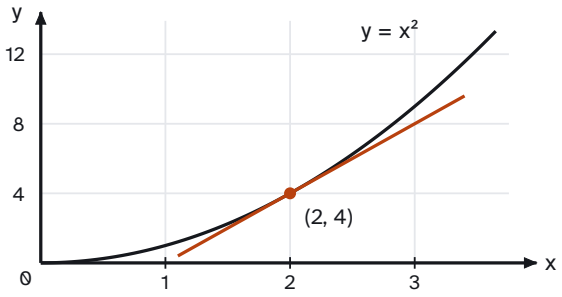
01 Idea

Differentiate a sum term by term. Constants disappear. The derivative gives the slope at a point, but the tangent line must also pass through the point on the original function. Rewrite roots and reciprocal powers before applying the power rule.

A derivative supplies a slope, not a whole line. Many parallel lines share that slope, so a tangent must also be anchored at its point on the original curve. Calculate the curve height from the original function and the slope from its derivative.

FIG. 04 / CALCULUS

A tangent measures the local rate of change



At x = 2

Slope = 2x = 4

Tangent equation:
y - 4 = 4(x - 2)

Takeaway: use coordinate changes for slope; the axes have different scales.

Figure 04. The tangent to $y = x^2$ at (2, 4) has slope 4 and equation y - 4 = 4(x - 2). Its plotted line touches the curve at the marked point. Read slope from coordinates, not the visual angle.

- The original function gives the tangent point.
- The derivative gives the local slope, and a point plus a finite slope determines the line.

02 Rule and symbols

Rule

$$\frac{d}{dx}x^n = nx^{n-1}$$

$$y - f(a) = f'(a)(x - a)$$

a input coordinate of tangent point / input unit

$f(a)$ output coordinate of tangent point / output unit

$f'(a)$ tangent slope / output unit per input unit

03 Method

1. Differentiate the original function term by term.
2. Substitute x = a into the derivative for the slope.
3. Substitute x = a into the original function for the point.
4. Use y - f(a) = f′(a)(x-a), then simplify only if requested.

A tangent-equation request needs both a point and a slope, even when differentiation itself is simple.

04 Worked example

For $f(x) = x^2 + 1$ at $x = 2$, the point is (2,5) and slope is 4.

01 For f(x) = x^2 + 1, f′(x) = 2x because the constant contributes zero.

02 At x = 2 the slope is 4, while the original function gives y = 5.

03 The tangent must therefore pass through (2,5) with slope 4.

04 Its equation is y - 5 = 4(x-2), which expands to y = 4x - 3.

Watch out

Do not use f'(a) as the y-coordinate.

Exam habit

Keep the unsimplified point-slope equation visible: it shows both required ingredients for method marks.

05 Your turn

Find the tangent to f(x)=x cubed at $x = 2$.

If you get stuck, uncover one hint at a time:

1. Evaluate x^3 at $x = 2$ to find the point on the original curve.
2. Differentiate x^3 and evaluate the derivative at the same input.
3. Use the point and slope in $y - y_0 = m(x - x_0)$, then check that the point satisfies your line.

06 Check your understanding

Does substituting x = 2 into f′ give the y coordinate of the tangent point?

Two points have the same finite derivative value. Their tangent lines must be:

1. Exactly the same line in every case
2. Parallel or coincident
3. Perpendicular

Answers and explanations: page 45.

Product and quotient rules

C1 / 03

- Differentiate a product using both changing factors.
- Preserve term order and the squared denominator in the quotient rule.

You need: Basic polynomial and sine derivatives; multiplication and fractions.

01 Idea

A product has two changing factors, so both contribute to the derivative. A quotient rule numerator must preserve its order; reversing it changes the sign. Do not differentiate numerator and denominator separately to differentiate a fraction.

When two factors change, a product changes through both of them. A quotient also has two changing parts, but its denominator contribution enters with a subtraction. Writing the full rule before simplifying preserves the separate contributions and protects the sign.

- The product rule includes one contribution from each changing factor.
- Reversing the quotient-rule subtraction reverses the final derivative sign.

02 Rule and symbols

Rule

$$(uv)' = u'v + uv'$$

$$(u/v)' = (u'v - uv')/v^2$$

u, v the two factor functions / problem dependent

u', v' their derivatives with respect to x / corresponding output per input unit

03 Method

1. Identify whether the expression is a product or quotient at the outermost level.
2. Name the component functions u and v and calculate their derivatives.
3. Insert all terms into the appropriate rule before simplifying.
4. For a quotient retain the original domain restriction and check the numerator subtraction order.

Inspect the outer operation: multiplication or division of functions signals the corresponding rule.

04 Worked example

For $y = x^2 \sin x$, derivative is $2x \sin x + x^2 \cos x$.

01 For y = x^2 sin x, choose u = x^2 and v = sin x.

02 Their derivatives are u′ = 2x and v′ = cos x.

03 The product rule keeps each factor once unchanged: u′v + uv′.

04 Thus y′ = 2x sin x + x^2 cos x; multiplying the two derivatives alone would miss both contributions.

Watch out

The quotient denominator is the square of the original denominator.

Exam habit

Write the entire quotient-rule numerator in brackets before expanding; the minus sign applies to the whole second product.

05 Your turn

Differentiate y=(x+1)/(x-1), for x not equal to 1.

If you get stuck, uncover one hint at a time:

1. Choose $u = x + 1$ and $v = x - 1$, then find both derivatives.
2. Form $(u'v - uv')/v^2$ with brackets around both numerator terms.
3. Simplify $(x - 1) - (x + 1)$ carefully and preserve $x \neq 1$.

06 Check your understanding

For y = (x+1)/(x-1), what is the derivative for x $\neq$ 1?

Why is separately differentiating top and bottom generally wrong for the derivative of u/v?

1. The denominator must always be constant
2. Fractions cannot be differentiated
3. It omits the complete effect of denominator change

Answers and explanations: page 45.

Chain rule and trigonometric functions

C1 / 04

- Identify inner and outer operations in a composite function.
- Include the inner derivative in trigonometric and power derivatives.

You need: Power rule; sine and cosine derivatives; radians.

01 Idea

A composite function has an outer operation acting on an inner expression. Differentiate the outside while preserving the inside, then multiply by the derivative of the inside. Trigonometric derivative formulas use radians.

A composite function is a machine inside a machine. The outer function responds to its inner input, while that inner input changes with the original variable. The chain rule multiplies these rates to describe the complete change.

- An outer derivative alone measures change per unit of inner expression.
- The inner derivative converts that rate into change per unit of the original variable.

02 Rule and symbols

Rule

$$\frac{d}{dx} f(g(x)) = f'(g(x))g'(x)$$

$$(\sin x)' = \cos x$$

$$(\cos x)' = -\sin x$$

$u = g(x)$ inner expression / problem dependent

$g'(x)$ inner derivative / inner-expression unit per input unit

03 Method

1. Identify the last operation applied to the expression: this is the outer function.
2. Differentiate that outer function while leaving the inner expression in place.
3. Multiply by the derivative of the inner expression.

4. Check the sign for cosine derivatives and use radians for trigonometric rules.

An operation surrounding a nontrivial expression, such as cosine surrounding a square, is a chain-rule cue.

04 Worked example

For $y = (3x + 1)^4$, derivative = $4(3x + 1)^3 \cdot 3 = 12(3x + 1)^3$.

01 In y = (3x+1)^4, the inner expression is 3x+1 and the outer operation is raising to the fourth power.

02 The outer derivative gives 4(3x+1)^3.

03 The inner derivative is 3, so it must multiply the outer result.

04 Combining the constants gives y′ = 12(3x+1)^3.

Watch out

Forgetting the inner derivative is the most common lost mark.

Exam habit

Before finishing, circle the inner expression and verify that its derivative appears as a factor.

05 Your turn

Differentiate y=cos(x^2).

If you get stuck, uncover one hint at a time:

1. Identify cosine as the outer function and x^2 as the inner expression.
2. Differentiate cosine while leaving x^2 inside, and retain the negative sign.
3. Multiply by the derivative of x^2. Check that both rates appear.

06 Check your understanding

What is the derivative of cos(x^2)?

In differentiating $\sin(7x)$, what does the extra factor 7 represent?

1. The rate of change of the inner expression
2. A second sine factor
3. A degrees-to-radians conversion

Answers and explanations: page 45.

Four more trigonometric derivatives

The supplied final asks for a cotangent derivative. Learn these rules together with sine, cosine and the chain rule.

- Use radians. Here u is a function of x, and u' means its derivative.
- Apply the outside rule, then multiply by the inside derivative.
- Use a rule only where the original function is defined. Remember: $\sec u = 1/\cos u$, $\csc u = 1/\sin u$, and $\cot u = \cos u/\sin u$.
- $\dfrac{d}{dx}\tan u = \sec^2 u\, u'$
- $\dfrac{d}{dx}\cot u = -\csc^2 u\, u'$
- $\dfrac{d}{dx}\sec u = \sec u \tan u\, u'$
- $\dfrac{d}{dx}\csc u = -\csc u \cot u\, u'$

Differentiate $y = \cot(5x)$.

1. The inside is $u = 5x$, so $u' = 5$.
2. The cotangent rule gives $y' = -\csc^2(5x) \times 5$.

$y' = -5\csc^2(5x)$. The minus sign and the factor 5 both matter.

Implicit differentiation

C1 / 05

- Differentiate y terms while treating y as a function of x.
- Collect and isolate all terms containing y′.

You need: Chain and product rules; algebraic rearrangement.

01 Idea

When x and y appear in one equation, y is treated as a function of x. Differentiating a y term therefore introduces y'. Collect all y-prime terms and solve algebraically. This can describe curves that are not single-valued functions globally.

An implicit equation links two changing coordinates without isolating one globally. As one coordinate changes, the other adjusts to stay on the curve. Differentiation records this linked movement, so differentiating a function of the dependent coordinate requires its own rate of change.

- A y-expression is a composite function of x, so chain-rule factors appear.
- Collecting derivative terms reveals a shared factor that can be isolated to find the slope.

02 Rule and symbols

Rule

$$\frac{d}{dx}(y^2) = 2yy'$$

$$\frac{d}{dx}(xy) = y + xy'$$

y' dy/dx, the local slope of the curve / y unit per x unit

03 Method

1. Differentiate both sides with respect to x.
2. Attach y′ when differentiating a function of y, and use the product rule for xy.
3. Move all terms containing y′ to one side.
4. Factor out y′, divide by its coefficient and state where that division is valid.

Use implicit differentiation when x and y are mixed and explicit rearrangement would be awkward or lose branches.

04 Worked example

From $x^2 + y^2 = 25$, differentiation gives $2x + 2yy' = 0$.

01 The curve $x^2 + y^2 = 25$ need not be solved globally for a single y value.

02 Differentiating x^2 gives 2x; differentiating y^2 gives 2y y′ by the chain rule.

03 The constant 25 differentiates to zero, so 2x + 2y y′ = 0.

04 Rearranging gives y′ = -x/y wherever $y \neq 0$; division by y requires this restriction.

Watch out

Do not treat y as a constant or omit the product rule on xy.

Exam habit

For an xy term write y + xy′, not simply y or xy′; both factors depend on x.

05 Your turn

Find dy/dx for xy+y squared=6.

If you get stuck, uncover one hint at a time:

1. Differentiate both sides of $xy + y^2 = 6$, using the product rule for xy.
2. Include y' when differentiating y^2, and collect all derivative terms.
3. Factor out y', move the remaining term and divide by its coefficient where nonzero.

06 Check your understanding

Why does the derivative of y^2 contain $y\prime$?

When y depends on x, the derivative of xy is:

1. xy'
2. $y + xy'$
3. y

Answers and explanations: page 45.

Logarithmic, exponential and inverse-trig derivatives

C1 / 06

- Differentiate exponential and logarithmic composites.
- Use logarithmic differentiation for a positive variable base and exponent.

You need: Chain rule; natural logarithms; product rule; positive-domain restrictions.

01 Idea

Exponential functions reproduce themselves up to an inner derivative. Natural logarithms give a reciprocal factor. Logarithmic differentiation helps when variable powers or products are awkward. Inverse-trigonometric functions are different from reciprocal trigonometric functions.

A variable power changes through both its base and its exponent. A logarithm moves the exponent into a product, where familiar differentiation rules track those two changes. The final step restores the original function after differentiating the logarithmic equation.

- Logarithms turn a positive power expression into a product involving its exponent.
- The derivative of the logarithm includes the original derivative divided by the original function, so that function must be restored.

02 Rule and symbols

Rule

$$(e^u)' = e^u u'$$

$$(\ln|u|)' = u'/u$$

$$(\arctan x)' = 1/(1+x^2)$$

e base of the natural exponential / dimensionless

$\ln u$ natural logarithm of a positive dimensionless argument / dimensionless

03 Method

1. Identify the outer function and check its real domain.
2. For a logarithm use inner derivative divided by the inner expression.
3. For an exponential retain the exponential and multiply by the inner derivative.
4. For y = x^x with x > 0, take logs first, differentiate implicitly, then restore y.

A variable in both base and exponent suggests logarithmic differentiation, after checking a suitable positive domain.

04 Worked example

For $y = \ln(x^2 + 1)$, derivative is $\frac{2x}{x^2+1}$.

01 For ln(x^2+1), the inner derivative is 2x and the inner expression is always positive.

02 The logarithmic chain rule therefore gives 2x/(x^2+1).

03 For the separate function e^(3x), the exponential is retained unchanged.

04 Multiplication by the inner derivative 3 gives 3e^(3x).

Watch out

The derivative of ln u is u prime/u, not 1/ln u.

Exam habit

The power rule nx^(n-1) assumes a constant exponent; x^x needs another method.

05 Your turn

Differentiate y=x to the power x for x greater than zero.

If you get stuck, uncover one hint at a time:

1. For $x > 0$, set $y = x^x$ and take natural logarithms.

2. Rewrite the right side as $x \ln x$; use the product rule while the left derivative is y'/y.
3. Multiply by y, then replace it with the original x^x.

06 Check your understanding

For x > 0, what is d(x^x)/dx?

Why is the constant-exponent power rule insufficient for x^x?

1. The expression is constant
2. Logarithms are invalid for positive x
3. The exponent changes as well as the base

Answers and explanations: page 45.

Other exponential, logarithm and inverse-trig rules

The supplied final includes 3^x and $\log_3 x$. Their derivatives need a factor involving $\ln 3$.

- The base a is a fixed positive number, with $a \neq 1$. For the logarithm, require $u > 0$.
- For the displayed arcsine and arccosine derivatives, require $|u| < 1$. Angles are in radians.
- Find the inside derivative u' first. Inverse sine means $\arcsin$; it is different from $1/\sin$.
- $\dfrac{d}{dx} a^u = a^u \ln(a)\, u'$
- $\dfrac{d}{dx} \log_a u = \dfrac{u'}{u \ln a}$
- $\dfrac{d}{dx} \arcsin u = \dfrac{u'}{\sqrt{1-u^2}}$

- $\dfrac{d}{dx}\arccos u = -\dfrac{u'}{\sqrt{1-u^2}}$

Differentiate 3^x and $\log_3 x$.

1. In both cases the inside expression is $u = x$, so $u' = 1$.
2. For the exponential, multiply by $\ln 3$. For the logarithm, divide by $\ln 3$.

$(3^x)' = 3^x \ln 3$, and $(\log_3 x)' = 1/(x \ln 3)$ for $x > 0$.

L Hopital and indeterminate forms

C1 / 07

- Check for a permitted indeterminate quotient before using L Hopital.
- Evaluate a simple limit by differentiating numerator and denominator.

You need: Limits; exponential and trigonometric derivatives; chain rule.

01 Idea

L Hopital rule can evaluate a quotient limit with the form $\frac{0}{0}$ or $\frac{\infty}{\infty}$ under its differentiability and limit conditions. Differentiate numerator and denominator separately for this limit procedure; this is not the quotient rule. Other indeterminate forms must first be rewritten.

An indeterminate quotient describes competition between quantities approaching zero or growing without bound. When its conditions hold, L Hopital compares their rates of change. It is a limit procedure, so it does not replace the quotient rule for the derivative of a fraction.

- The initial limiting form must be checked before applying the rule.
- Differentiating a numerator containing an inner expression still requires the chain rule.

02 Rule and symbols

Rule

$$\lim \frac{f(x)}{g(x)} = \lim \frac{f'(x)}{g'(x)}$$

when the rule applies

$0/0$ indeterminate limiting form, not an arithmetic value / not a measured quantity

∞/∞ indeterminate quotient form / not a measured quantity

Use this when:

The numerator and denominator satisfy the differentiability and denominator-derivative conditions near the limiting point.

03 Method

1. Substitute the limiting input to identify the quotient form.
2. Confirm 0/0 or infinity/infinity and the rule conditions before differentiating.
3. Differentiate numerator and denominator separately, not with the quotient rule.
4. Evaluate the new limit; if still indeterminate, check conditions again before repeating.

Check the quotient form first; an ordinary fraction alone is not a reason to use L Hopital.

04 Worked example

As x approaches zero, $\frac{e^x-1}{x}$ has form $\frac{0}{0}$.

01 As x approaches zero, e^x - 1 approaches zero and x approaches zero.

02 The quotient therefore has the required 0/0 form.

03 Differentiate numerator and denominator separately to obtain e^x/1.

04 Its limit is e^0 = 1, so the original limit is 1.

Watch out

Do not use the rule on a quotient whose substitution gives an ordinary finite value.

Exam habit

Show the initial indeterminate form; applying the rule without checking its conditions can invalidate the method.

05 Your turn

Evaluate the limit of sin(3x)/x as x approaches zero.

If you get stuck, uncover one hint at a time:

1. Check the numerator and denominator as $x \to 0$ and identify the form.
2. Differentiate $\sin(3x)$ with its inner derivative and differentiate the denominator separately.
3. Evaluate the resulting derivative quotient at the limiting input; do not use a quotient-rule numerator.

06 Check your understanding

What is the limit of sin(3x)/x as x approaches zero?

A numerator approaches 2 and its denominator approaches 4. The appropriate first method is:

1. Use the ordinary quotient limit
2. Apply L Hopital because it is a fraction
3. Declare the form indeterminate

Answers and explanations: page 45.

Increasing, decreasing and concavity

C1 / 08

- Use a first-derivative sign chart to find monotonic intervals.
- Distinguish a stationary point from a proven extremum or inflection.

You need: Factoring quadratics; sign tests; first and second derivatives.

01 Idea

The sign of the first derivative controls increasing and decreasing intervals. The sign of the second controls concavity. A critical number lies in the domain where the first derivative is zero or undefined. An inflection point requires a concavity change, not merely a zero second derivative.

The derivative records the local slope of the graph. A negative derivative makes the graph fall as input increases; a positive derivative makes it rise. A change from falling to rising identifies a minimum, whereas a zero derivative without a sign change may not be an extremum.

- A zero derivative gives a candidate rather than a complete extremum proof.
- Signs on both sides tell whether increasing or decreasing behaviour actually changes.

02 Rule and symbols

Rule

$$f' > 0 : \text{ increasing;}$$
$$f'' > 0 : \text{ concave up}$$

f' first derivative, controlling increase or decrease / output per input unit

f'' second derivative, controlling concavity / output per input unit squared

03 Method

1. Find where f′ is zero or undefined within the original domain.
2. Split the domain at those critical numbers and test the sign on each interval.
3. Use sign changes to classify local maxima and minima.

4. Use f″ signs separately for concavity and require a concavity change for an inflection point.

Use a derivative sign chart for monotonic intervals and local extrema; use a separate concavity test for inflections.

04 Worked example

For $f = x^3 - 3x$, derivative is $3(x-1)(x+1)$.

01 For f(x) = x³ - 3x, f′ = 3x² - 3 = 3(x-1)(x+1).

02 The critical numbers are -1 and 1.

03 Testing x = -2, 0 and 2 gives derivative signs positive, negative and positive.

04 The function increases on (-∞,-1) and (1,∞), and decreases on (-1,1); the sign changes identify a local maximum at -1 and minimum at 1.

Watch out

A derivative equal to zero does not automatically mean a maximum.

Exam habit

Give open intervals for increasing and decreasing behaviour, then state extrema as points if asked.

05 Your turn

For $f(x) = x^2$-4x, find increasing and decreasing intervals and the minimum point.

If you get stuck, uncover one hint at a time:

1. Differentiate $x^2 - 4x$ and find where its derivative is zero.
2. Test derivative signs using one input on each side of that critical number.
3. Report the resulting intervals and evaluate the original function at the critical input to give the minimum point.

06 Check your understanding

Is f″(c) = 0 alone enough to prove an inflection point?

A first derivative changes from negative to positive at an interior critical point. This gives:

1. An inflection point guaranteed
2. A local minimum
3. A local maximum

Answers and explanations: page 45.

Graphing, asymptotes and absolute extrema

C1 / 09

- Find absolute extrema by comparing all candidates on a closed interval.
- Distinguish a removable hole from a vertical asymptote.

You need: Domain restrictions; critical numbers; evaluating a function at endpoints.

01 Idea

Graphing combines domain, intercepts, derivative signs and limiting behaviour. A cancelled factor can create a hole rather than an asymptote. On a closed interval a continuous function attains absolute extrema; compare endpoint and interior critical values.

An absolute extremum compares the entire allowed graph. On a closed interval, an endpoint can be highest or lowest without having a horizontal tangent. Comparing a short list of original function values turns this global problem into a finite check.

- Interior critical numbers supply possible internal extrema.
- Endpoints must also be considered because the allowed domain ends there.

02 Rule and symbols

Rule

Candidates = endpoints plus interior critical numbers

$[a, b]$ closed interval including both endpoints / input unit

$f(c)$ function value at a candidate point / output unit

03 Method

1. Check the domain and whether the function is continuous on the stated closed interval.
2. Find all interior critical numbers.
3. Evaluate the original function at those numbers and at both endpoints.

4. Compare the values and report each extremum value together with where it occurs.

Absolute maximum or minimum on a closed interval calls for endpoint-plus-critical-value comparison.

04 Worked example

For x^2 on $[-1, 2]$, candidates are -1,0,2.

01 For f(x) = x^2 on [-1,2], the derivative 2x is zero at x = 0.

02 The candidate inputs are the two endpoints -1 and 2 and the interior critical number 0.

03 The original function values are f(-1)=1, f(0)=0 and f(2)=4.

04 The smallest is 0 at x=0; the largest is 4 at x=2.

Watch out

Do not omit endpoints or compare derivative values instead of function values.

Exam habit

Use the original function for the comparison table; comparing derivative values does not locate the highest graph point.

05 Your turn

Find absolute maximum and minimum of $f(x) = x^2$-2x on [0,3].

If you get stuck, uncover one hint at a time:

1. Find the derivative of $x^2 - 2x$ and its critical input inside $[0, 3]$.
2. List that input and both endpoints as candidates.
3. Evaluate the original function at each candidate; report both the largest and smallest values and where each occurs.

06 Check your understanding

Can an endpoint be the absolute maximum even when its derivative is not zero?

A continuous function is strictly increasing on a closed interval. Its absolute maximum occurs:

1. Only where its derivative is zero
2. Nowhere because it keeps increasing
3. At the right endpoint

Answers and explanations: page 45.

Optimization and mean-value theorems

C1 / 10

- Translate a simple optimization constraint into one variable.
- Check mean-value-theorem assumptions and solve for the interior input.

You need: Area and perimeter; derivatives; continuity and differentiability.

01 Idea

Optimization translates a word problem into a one-variable function and an allowed domain. Rolle theorem requires equal endpoint values as well as continuity and interior differentiability. The mean-value theorem matches some instantaneous slope to the average slope.

The mean-value theorem links average change across an interval with instantaneous change somewhere inside. A secant joins endpoint heights, and at least one interior tangent must be parallel under the required conditions. The theorem matches slopes rather than automatically selecting a midpoint.

- Continuity and interior differentiability exclude breaks and corners that could defeat the guarantee.
- The matching input lies strictly inside the interval but need not be its midpoint.

02 Rule and symbols

Rule

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

$$a < c < b$$

x one rectangle side length / length unit

A rectangle area / square length unit

c interior input guaranteed by the mean-value theorem / input unit

03 Method

1. For optimization name the objective, write the constraint and identify the feasible domain.
2. Eliminate one variable, differentiate and find interior critical candidates.
3. Check endpoints or derivative signs to justify the optimum.
4. For a theorem question verify its assumptions before setting instantaneous slope equal to average slope.

Use the theorem when a function and interval are supplied and an interior slope-matching input is requested.

04 Worked example

A rectangle with perimeter 20 has area $x(10-x)$.

01 Perimeter 20 gives 2x + 2y = 20, so the other side is y = 10 - x.

02 Area becomes A(x) = x(10-x) with 0 < x < 10 for a nondegenerate rectangle.

03 Its derivative is 10 - 2x, which vanishes at x = 5.

04 The second derivative is -2, confirming a maximum; both sides are 5 and area is 25 square units.

Watch out

A candidate outside the physical domain is not a valid optimization answer.

Exam habit

A critical number alone is not a maximum proof; include a sign change, second derivative or candidate comparison.

05 Your turn

Use the mean-value theorem for $f(x)=x^2$ on [1,3] to find c.

If you get stuck, uncover one hint at a time:

1. Check that x^2 is continuous on $[1, 3]$ and differentiable inside.
2. Compute the endpoint secant slope from function values and interval length.
3. Set $f'(c)$ equal to that slope, solve and verify $1 < c < 3$.

06 Check your understanding

For f(x)=x^2 on [1,3], what c satisfies the mean-value theorem?

Must the input guaranteed by the mean-value theorem be the midpoint?

1. No; it solves the slope-matching equation
2. Yes; every function uses the midpoint
3. It must be an endpoint

Answers and explanations: page 45.

Antiderivatives and substitution

C1 / 11

- Choose a substitution whose differential matches the integrand.
- Return the antiderivative to x and include the integration constant.

You need: Chain rule; power integration; constant factors.

01 Idea

An indefinite integral represents a family of antiderivatives, so include an arbitrary constant. Substitution reverses the chain rule: select an inner expression whose derivative is present. Differentiate your answer as a fast correctness check.

Substitution recognizes the chain rule running backward. A temporary variable replaces an inner expression, while its differential accounts for a matching factor elsewhere. A leftover constant is harmless if tracked carefully rather than dropped.

- A useful substitution replaces the nested expression and its differential factor together.
- Differentiating the final result checks both the power and its multiplier.

02 Rule and symbols

Rule

$$\int x^n\,dx = x^{n+1}/(n+1) + C\ (n \neq -1)$$

$$\int dx/x = \ln|x| + C$$

u chosen inner expression / problem dependent

du differential of the inner expression / same unit as u

C arbitrary additive integration constant / antiderivative unit

03 Method

1. Choose an inner expression whose derivative is also present as a factor.
2. Write u and du and replace every part of the integral consistently.
3. Integrate in u, preserving any numerical factors.

4. Substitute back to x, add C and differentiate once to check.

Look for an inner expression accompanied by its derivative or a constant multiple of that derivative.

04 Worked example

For the integral of $2x(x^2 + 1)^3$, set $u = x^2 + 1$ and $du = 2x\,dx$.

01 For the integral of $2x(x^2+1)^3$, choose $u = x^2+1$.

02 Then $du = 2x\,dx$, so the complete integral becomes the integral of u^3 du.

03 The antiderivative is $u^4/4 + C$.

04 Substitute back to get $(x^2+1)^4/4 + C$; differentiation restores $2x(x^2+1)^3$.

Watch out

The power rule exception is n=-1; do not divide by zero.

Exam habit

Do not leave a mixture of u and x in the substituted integral; it hides missing differential factors.

05 Your turn

Evaluate the integral of $6x(x^2+2)$ squared with respect to x.

If you get stuck, uncover one hint at a time:

1. Choose $u = x^2 + 2$, calculate du and compare it with $6x\,dx$.
2. Rewrite the full integral in u, retaining any constant multiplier.
3. Integrate the power, return to x, include the arbitrary constant and differentiate to check.

06 Check your understanding

What is the integral of 6x(x^2+2)^2 dx?

Which substitution most directly simplifies $6x(x^2+2)^2$?

1. Rename the whole integrand without computing a differential
2. $u = x^2 + 2$
3. $u = 6x$

Answers and explanations: page 45.

Integrate sine, cosine and an exponential

These short rules reverse the chain rule. A sine integral appears in the supplied final.

- Here a and b are constants and $a \neq 0$. Trigonometric angles use radians.
- Divide by the coefficient a; differentiation would multiply by it.
- Add C for an indefinite integral. Differentiate your answer to check it.

- $\int \sin(ax+b)\,dx = -\dfrac{\cos(ax+b)}{a} + C$
- $\int \cos(ax+b)\,dx = \dfrac{\sin(ax+b)}{a} + C$
- $\int e^{ax+b}\,dx = \dfrac{e^{ax+b}}{a} + C$

Find $\int \sin(3x)\,dx$.

1. The sine antiderivative is negative cosine.
2. The inside derivative is 3, so divide by 3.
3. Check: $\dfrac{d}{dx}[-\cos(3x)/3] = \sin(3x)$.

$-\dfrac{1}{3}\cos(3x) + C$.

Definite integrals, signed area and the fundamental theorem

C1 / 12

- Distinguish signed integral from total geometrical area.
- Apply the fundamental theorem, including a variable-upper-limit chain factor.

You need: Antiderivatives; positive and negative regions; substituting bounds.

01 Idea

A definite integral gives signed accumulation: contributions below the axis are negative. Geometrical area requires positive contributions from each region. The fundamental theorem connects integration and differentiation; for a variable upper limit, include its derivative.

A definite integral combines signed contributions over fixed bounds. An antiderivative provides a shortcut through its change between endpoints. If the integrand is nonnegative throughout, the signed accumulation is also geometric area; otherwise positive and negative regions can cancel.

- The fundamental theorem requires upper antiderivative value minus lower value.
- Substitution changes the endpoint labels unless the antiderivative is returned to the original variable before evaluation.

02 Rule and symbols

Rule

$$\int_a^b f(x)\,dx = F(b) - F(a)$$

$$\frac{d}{dx}\int_a^{g(x)} f(t)\,dt = f(g(x))g'(x)$$

a, b lower and upper integration bounds / input unit

F an antiderivative of f / f unit times input unit

t dummy integration variable / input unit

03 Method

1. Identify whether the question asks for signed accumulation or total area.
2. Find an antiderivative and calculate upper value minus lower value.
3. For total area split at sign changes and make each region contribution positive.
4. For differentiation with upper bound g(x), evaluate f at g(x) and multiply by g′(x).

With fixed limits, keep endpoint values consistent with the integration variable and evaluate their difference.

04 Worked example

The $\int_{-1}^{1} x\,dx$ is zero by cancellation, but total geometrical area is 1.

01 For y = x from -1 to 1, the region below the axis contributes negative signed area.

02 An antiderivative is $x^2/2$; evaluation at 1 and -1 gives 1/2 - 1/2 = 0.

03 Geometrically each half is a triangle with area 1/2.

04 Total geometrical area is 1/2 + 1/2 = 1, while the signed integral remains zero.

Watch out

Never keep x-limits on an integral that has been fully changed to u.

Exam habit

For a definite integral evaluate both bounds; no arbitrary +C is needed in the final numerical answer.

05 Your turn

Evaluate the integral from 0 to 1 of 2x(x^2+1) dx.

If you get stuck, uncover one hint at a time:

1. Notice $x^2 + 1$ and its derivative factor $2x\,dx$.
2. Substitute with transformed bounds, or find an antiderivative and return it to x.
3. Evaluate upper minus lower between 0 and 1. Since the integrand is nonnegative there, a negative result would indicate an error.

06 Check your understanding

Does a zero definite integral prove the function is zero throughout the interval?

After changing variable in a definite integral, what should happen to the bounds?

1. Always keep the old bound numbers with the new variable
2. Remove them permanently
3. Transform them, or return to the original variable before evaluation

Answers and explanations: page 45.

Read a Riemann sum as rectangles

The course syllabus includes sigma notation and area as a limit. The symbol $\sum$ simply means add the terms.

- Split the interval from a to b into n equal widths.
- Choose a sample point x_i^* in each small interval. Its function value is that rectangle's signed height.
- Add height times width. More, narrower rectangles approach the definite integral when the function is continuous.
- Below-axis heights contribute negative signed area.
- $\Delta x = \dfrac{b-a}{n}$
- rectangle sum $= \sum_{i=1}^{n} f(x_i^*)\,\Delta x$

Use two right-end rectangles to estimate $\int_0^1 x\,dx$.

1. Each width is $\Delta x = (1-0)/2 = 1/2$.
2. The right endpoints are $1/2$ and 1, giving heights $1/2$ and 1.
3. The estimate is $(1/2)(1/2) + (1)(1/2) = 3/4$.

Estimate: $3/4$. Exact integral: $1/2$. The right-end estimate is too large because the function increases.

Lesson review and answers

Limits, continuity and the derivative definition

Return to lesson: page 6.

Your turn: It would produce division by zero; the limit concerns nonzero h approaching zero.

Concept check: 1. Correct. Simplification is valid at nonzero increments, and those nearby values determine the limit.

Written method: [3(x+h) squared -3x squared]/h = [6xh+3h squared]/h = 6x+3h. Taking the limit gives 6x. One mark each for setup, expansion, cancellation and limit.

Basic derivatives and tangent lines

Return to lesson: page 9.

Your turn: No. It gives the slope; use f(2) for the y coordinate.

Concept check: 2. Correct. Equal slopes give parallel lines, which can coincide in a special case.

Written method: f(2)=8; derivative=3x squared; slope at 2 is 12. Tangent y-8=12(x-2), equivalently y=12x-16. One mark per step.

Product and quotient rules

Return to lesson: page 12.

Your turn: -2/(x-1)^2: the numerator is (x-1) - (x+1) = -2.

Concept check: 3. Correct. The quotient rule accounts for both changes and includes the squared denominator.

Written method: Derivative = [(x-1)-(x+1)]/(x-1) squared = -2/(x-1) squared. Award marks for quotient rule, numerator order, denominator square and simplification.

Chain rule and trigonometric functions

Return to lesson: page 15.

Your turn: -2x sin(x^2), combining the cosine derivative with the inner derivative 2x.

Concept check: 1. Correct. The inner expression changes seven units per unit increase in x.

Written method: Outer derivative is -sin(x^2); inner derivative is 2x. Multiply to

obtain -2x sin(x^2). Credit both layers, sign and product.

Implicit differentiation

Return to lesson: page 19.

Your turn: Because y depends on x; the chain rule multiplies 2y by dy/dx.

Concept check: 2. Correct. Both factors change, so the product rule gives both terms.

Written method: Differentiate: y+x y'+2y y'=0. Collect: (x+2y)y'=-y. Hence y'=-y/(x+2y), where the denominator is nonzero. Four marks across these steps.

Logarithmic, exponential and inverse-trig derivatives

Return to lesson: page 22.

Your turn: x^x(ln x + 1). Take ln y = x ln x, then y′/y = ln x + 1.

Concept check: 3. Correct. Treating the exponent as fixed misses part of the change.

Written method: Take logarithms: ln y=x ln x. Differentiate: y'/y=ln x+1. Multiply by y: y'=x to the x times (ln x+1). Four marks for these stages.

L Hopital and indeterminate forms

Return to lesson: page 26.

Your turn: 3, since the derivative quotient is 3 cos(3x)/1 and cos 0 = 1.

Concept check: 1. Correct. A nonzero finite denominator limit gives a directly determined quotient.

Written method: The form is $\frac{0}{0}$. L Hopital gives limit 3 cos(3x)/1 = 3. Alternatively use the standard sine limit. Credit form, derivative, evaluation and answer.

Increasing, decreasing and concavity

Return to lesson: page 29.

Your turn: No. Concavity must change across c.

Concept check: 2. Correct. The graph falls toward the point and rises away.

Written method: Derivative = 2x-4, zero at 2. It is negative for x<2 and positive for x>2. Minimum is (2,-4). One mark each for derivative, intervals and point.

Graphing, asymptotes and absolute extrema

Return to lesson: page 32.

Your turn: Yes. Absolute extrema on a closed interval must include endpoint checks.

Concept check: 3. Correct. The last allowed input has the largest output.

Written method: Critical point x=1. Values: f(0)=0, f(1)=-1, f(3)=3. Minimum -1 at x=1; maximum 3 at $x = 3$. Credit candidates, evaluations and both extrema.

Optimization and mean-value theorems

Return to lesson: page 35.

Your turn: c=2: the average slope is (9-1)/(3-1)=4 and f′(c)=2c=4.

Concept check: 1. Correct. A midpoint works for some functions but is not the general theorem conclusion.

Written method: The polynomial is continuous and differentiable. Average slope = (9-1)/(3-1)=4. Solve 2c=4 to get c=2, which is inside (1,3). Four marks for hypotheses, slope, solution and interval check.

Antiderivatives and substitution

Return to lesson: page 38.

Your turn: $(x^2+2)^3 + C$. With $u=x^2+2$, the integral becomes 3 times the integral of u^2 du.

Concept check: 2. Correct. Its differential matches the remaining x factor up to a constant.

Written method: Set $u=x^2+2$, $du = 2x\,dx$. Integral becomes 3 times integral u squared du = u cubed+C = (x^2+2) cubed+C. Credit substitution, factor, integration and constant.

Definite integrals, signed area and the fundamental theorem

Return to lesson: page 41.

Your turn: No. Positive and negative contributions can cancel.

Concept check: 3. Correct. The endpoints must still describe the same original inputs.

Written method: Set $u = x^2 + 1$; limits become 1 and 2. Integral of u du from 1 to 2 = (4-1)/2 = 3/2. Credit substitution, new limits, antiderivative and subtraction.

Calculus I / Test A

30 marks / Suggested time: 30 minutes. MCQs: 1 mark each. Written problems: 4 marks each. Attempt without notes.

Part I / Choose one answer

1. The derivative of x^2 at $x = 3$ is:

   A. 3

   B. 6

   C. 9

   D. 0

2. The derivative of a constant 7 is:

   A. 7

   B. 1

   C. x

   D. 0

3. The derivative of xe^x is:

   A. $e^x(1 + x)$

   B. 0

   C. e^x

   D. xe^x

4. The derivative of $\sin(5x)$ is:

   A. $-5\sin(5x)$

   B. $\sin(5)$

   C. $\cos(5x)$

   D. $5\cos(5x)$

5. For $x^2 + y^2 = 25$, the slope at (3,4) is:

   A. 0

   B. $\frac{3}{4}$

   C. $-\frac{3}{4}$

D. $\frac{4}{3}$

6. The derivative of $\arctan x$ is:

A. $\arctan(1)$

B. $\frac{1}{1+x^2}$

C. $\frac{1}{\cos^2 x}$

D. $-\frac{1}{1+x^2}$

Part II / Show your reasoning

1. Use the definition to find the derivative of f(x)=3x squared.

2. Find the tangent to f(x)=x cubed at $x = 2$.

3. Differentiate y=(x+1)/(x-1), for x not equal to 1.

4. Differentiate y=cos(x^2).

5. Find dy/dx for xy+y squared=6.

6. Differentiate y=x to the power x for x greater than zero.

Calculus I / Test B

30 marks / Suggested time: 30 minutes. MCQs: 1 mark each. Written problems: 4 marks each. Attempt without notes.

Part I / Choose one answer

1. Before applying L Hopital directly, the quotient should have form:

   A. $\frac{0}{0}$ or $\frac{\infty}{\infty}$

   B. $\frac{1}{1}$ only

   C. $\frac{0}{5}$ only

   D. Any fraction

2. If f' is negative on an interval, f is:

   A. Decreasing

   B. Constant

   C. Always concave down

   D. Increasing

3. To find absolute extrema on a closed interval, check:

   A. Only where f' is zero

   B. Only endpoints

   C. Endpoints and interior critical numbers

   D. Only where f is zero

4. Rolle theorem additionally requires:

   A. An infinite interval

   B. Equal endpoint function values

   C. A discontinuity

   D. A nonzero derivative everywhere

5. An indefinite integral must generally include:

   A. Only a numerical answer

   B. A fixed upper limit

   C. No variable

D. An arbitrary constant

6. The $\int_{-2}^{2} x\,dx$ equals:

A. 8

B. -4

C. 0

D. 4

Part II / Show your reasoning

1. Evaluate the limit of sin(3x)/x as x approaches zero.

2. For $f(x) = x^2$-4x, find increasing and decreasing intervals and the minimum point.

3. Find absolute maximum and minimum of $f(x) = x^2$-2x on [0,3].

4. Use the mean-value theorem for $f(x) = x^2$ on [1,3] to find c.

5. Evaluate the integral of 6x(x^2+2) squared with respect to x.

6. Evaluate the integral from 0 to 1 of 2x(x^2+1) dx.

Calculus I / Test A / Answers

Suggested marking points are for self-study; your lecturer may use a different scheme.

Question 1

MCQ: B. The derivative is 2x, so its value at 3 is 6.

Written solution: [3(x+h) squared -3x squared]/h = [6xh+3h squared]/h = 6x+3h. Taking the limit gives 6x. One mark each for setup, expansion, cancellation and limit.

Return to lesson: page 6.

Question 2

MCQ: D. A constant function has zero rate of change.

Written solution: f(2)=8; derivative=3x squared; slope at 2 is 12. Tangent y-8=12(x-2), equivalently y=12x-16. One mark per step.

Return to lesson: page 9.

Question 3

MCQ: A. The product rule differentiates each factor in turn.

Written solution: Derivative = [(x-1)-(x+1)]/(x-1) squared = -2/(x-1) squared. Award marks for quotient rule, numerator order, denominator square and simplification.

Return to lesson: page 12.

Question 4

MCQ: D. The inner derivative contributes a factor of 5.

Written solution: Outer derivative is -sin(x^2); inner derivative is 2x. Multiply to obtain -2x sin(x^2). Credit both layers, sign and product.

Return to lesson: page 15.

Question 5

MCQ: C. Implicit differentiation gives slope $-\frac{x}{y}$.

Written solution: Differentiate: y+x y'+2y y'=0. Collect: (x+2y)y'=-y. Hence y'=-

y/(x+2y), where the denominator is nonzero. Four marks across these steps.

Return to lesson: page 19.

Question 6

MCQ: B. This is the standard inverse-tangent derivative.

Written solution: Take logarithms: ln y=x ln x. Differentiate: y'/y=ln x+1. Multiply by y: y'=x to the x times (ln x+1). Four marks for these stages.

Return to lesson: page 22.

Calculus I / Test B / Answers

Suggested marking points are for self-study; your lecturer may use a different scheme.

Question 1

MCQ: A. The rule concerns specified indeterminate quotient forms.

Written solution: The form is $\frac{0}{0}$. L Hopital gives limit 3 cos(3x)/1 = 3. Alternatively use the standard sine limit. Credit form, derivative, evaluation and answer.

Return to lesson: page 26.

Question 2

MCQ: A. Negative slope means the function decreases as x increases.

Written solution: Derivative = 2x-4, zero at 2. It is negative for x<2 and positive for x>2. Minimum is (2,-4). One mark each for derivative, intervals and point.

Return to lesson: page 29.

Question 3

MCQ: C. The largest or smallest value may occur at either kind of candidate.

Written solution: Critical point x=1. Values: f(0)=0, f(1)=-1, f(3)=3. Minimum -1 at x=1; maximum 3 at $x = 3$. Credit candidates, evaluations and both extrema.

Return to lesson: page 32.

Question 4

MCQ: B. Equal endpoint heights make the average slope zero.

Written solution: The polynomial is continuous and differentiable. Average slope = (9-1)/(3-1)=4. Solve 2c=4 to get c=2, which is inside (1,3). Four marks for hypotheses, slope, solution and interval check.

Return to lesson: page 35.

Question 5

MCQ: D. Differentiation loses constants, so all antiderivatives differ by a constant on an interval.

Written solution: Set u=x^2+2, $du = 2x\,dx$. Integral becomes 3 times integral u squared du = u cubed+C = (x^2+2) cubed+C. Credit substitution, factor, integration and constant.

Return to lesson: page 38.

Question 6

MCQ: C. The odd integrand has equal and opposite signed areas.

Written solution: Set $u = x^2 + 1$; limits become 1 and 2. Integral of u du from 1 to 2 = (4-1)/2 = 3/2. Credit substitution, new limits, antiderivative and subtraction.

Return to lesson: page 41.

Formula sheet

Limits, continuity and the derivative definition p.6

Rule

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

Basic derivatives and tangent lines p.9

Rule

$$\frac{d}{dx} x^n = n x^{n-1}$$

$$y - f(a) = f'(a)(x - a)$$

Product and quotient rules p.12

Rule

$$(uv)' = u'v + uv'$$

$$(u/v)' = (u'v - uv')/v^2$$

Chain rule and trigonometric functions p.15

Rule

$$\frac{d}{dx} f(g(x)) = f'(g(x)) g'(x)$$

$$(\sin x)' = \cos x$$

$$(\cos x)' = -\sin x$$

Implicit differentiation p.19

Rule

$$\frac{d}{dx}(y^2) = 2yy'$$

$$\frac{d}{dx}(xy) = y + xy'$$

Logarithmic, exponential and inverse-trig derivatives p.22

Rule

$$(e^u)' = e^u u'$$

$$(\ln |u|)' = u'/u$$

$$(\arctan x)' = 1/(1 + x^2)$$

L Hopital and indeterminate forms p.26

Rule

$$\lim \frac{f(x)}{g(x)} = \lim \frac{f'(x)}{g'(x)}$$

when the rule applies

Increasing, decreasing and concavity p.29

Rule

$$f' > 0 : \text{ increasing;}$$
$$f'' > 0 : \text{ concave up}$$

Graphing, asymptotes and absolute extrema p.32

Rule

Candidates = endpoints plus interior critical numbers

Optimization and mean-value theorems p.35

Rule

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$
$$a < c < b$$

Antiderivatives and substitution p.38

Rule

$$\int x^n \, dx = x^{n+1}/(n+1) + C \ (n \neq -1)$$
$$\int dx/x = \ln |x| + C$$

Definite integrals, signed area and the fundamental theorem p.41

Rule

$$\int_a^b f(x) \, dx = F(b) - F(a)$$
$$\frac{d}{dx} \int_a^{g(x)} f(t) \, dt = f(g(x))g'(x)$$

Symbol glossary

Symbols are listed with their lesson context. A symbol may represent different quantities in different rules.

Limits, continuity and the derivative definition

h small nonzero input increment before the limit / same unit as x

$f'(x)$ instantaneous rate of change / output unit per input unit

Basic derivatives and tangent lines

a input coordinate of tangent point / input unit

$f(a)$ output coordinate of tangent point / output unit

$f'(a)$ tangent slope / output unit per input unit

Product and quotient rules

u, v the two factor functions / problem dependent

u', v' their derivatives with respect to x / corresponding output per input unit

Chain rule and trigonometric functions

$u = g(x)$ inner expression / problem dependent

$g'(x)$ inner derivative / inner-expression unit per input unit

Implicit differentiation

y' dy/dx, the local slope of the curve / y unit per x unit

Logarithmic, exponential and inverse-trig derivatives

e base of the natural exponential / dimensionless

$\ln u$ natural logarithm of a positive dimensionless argument / dimensionless

L Hopital and indeterminate forms

$0/0$ indeterminate limiting form, not an arithmetic value / not a measured quantity

∞/∞ indeterminate quotient form / not a measured quantity

Increasing, decreasing and concavity

f' first derivative, controlling increase or decrease / output per input unit

f'' second derivative, controlling concavity / output per input unit squared

Graphing, asymptotes and absolute extrema

$[a, b]$ closed interval including both endpoints / input unit

$f(c)$ function value at a candidate point / output unit

Optimization and mean-value theorems

x one rectangle side length / length unit

A rectangle area / square length unit

c interior input guaranteed by the mean-value theorem / input unit

Antiderivatives and substitution

u chosen inner expression / problem dependent

du differential of the inner expression / same unit as u

C arbitrary additive integration constant / antiderivative unit

Definite integrals, signed area and the fundamental theorem

a, b lower and upper integration bounds / input unit

F an antiderivative of f / f unit times input unit

t dummy integration variable / input unit

Source and coverage notes

MATH 203: derivative rules, implicit/logarithmic/exponential derivatives, L Hopital, graphing and extrema, optimization, Rolle and mean-value theorems, integration and the fundamental theorem. The full lecture syllabus is the main source.

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The source audit inventories 159 non-programming files; 95 yielded substantial extracted text. Scanned diagrams and handwritten or image-only material were not fully transcribed. Check lecturer-specific derivations and conventions. The practice tests are original study material, not predictions of an examination.

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